pling frequency and 'n' is an integer called time index. Sample value $f(n)$ is an approximation of the signal amplitude at time instant $t = nT$.

Understanding the speech signal

Record vowel sound 'aa' using the computer's microphone and save it as a wav file. Select sampling frequency as 10kHz. You may use audio processing software like Praat, Audacity, Goldwave or Wavesurfer to record the signal in wav format at the required sampling frequency.

The waveform of the signal, which is a plot of the amplitude of the speech signal for each sample instant, looks like Fig. 1. The horizontal axis is time units in samples and the vertical axis is amplitude of the corresponding samples. If you record sound 'as' in which consonant sound 's' follows the vowel sound 'a,' and plot the signal, the waveform may look like Fig. 2.

On close examination of Fig. 1, you can see some repeating pattern in it. A zoomed version of Fig. 1 showing samples in the range 1000 to 1500 is given in Fig. 3. If you plot samples from 6000 to 6500 in Fig. 2, you get Fig. 4. Obviously, the waveform in Fig. 4 has no periodicity and it appears noise-like. In the waveform of vowel-consonant sound 'as,' you can see that the speech signal properties transform gradually from a nearly periodic signal (samples 2000 to 5000) to a noise-like signal (samples 5000 to 10000).

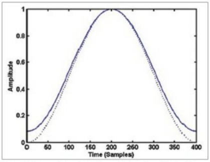

Fig. 5: Hamming (solid) and Hanning (dotted) windows of 400 samples

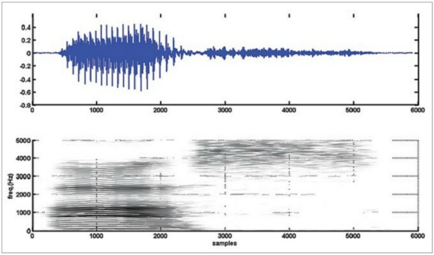

Fig. 6: Narrowband spectrogram of vowel-consonant sound 'as'

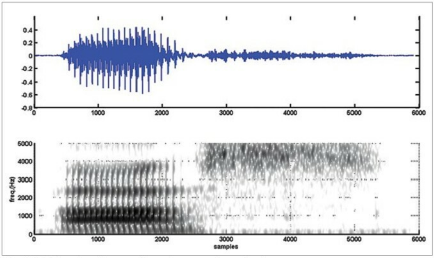

Fig. 7: Wideband spectrogram of vowel-consonant sound 'as'

Frequency-domain analysis of the speech signal

Waveform is a representation of the speech signal. It is a visualisation of the signal in time domain. This representation is almost silent on the frequency contents and the frequency distribution of energy in the speech signal. To get a frequency-domain representation, you need to take Fourier transform of the speech signal. Since speech signal has time-varying properties, the transformation from time-domain to frequency-domain also needs to be done in a time-dependent manner. In other words, you need to take small frames at different points along the time axis, take Fourier transform of the short-duration frames, and then proceed along the time axis towards the end of the utterance. The process is called short-time Fourier transform (STFT). Steps involved in STFT computation are:

1. Select a short-duration frame of the speech signal by windowing
2. Compute Fourier transform of the selected duration
3. Shift the window along the time axis to select the neighbouring frame
4. Repeat step 2 until you reach the end of the speech signal

To select a short-duration frame of speech, normally a window function with gradually rising and falling property is used. Commonly used window functions in speech processing are Hamming and Hanning windows. A Hamming window with 'N' points is mathematically represented by:

$$h_m(n) = 0.54 - 0.46\cos\frac{2\pi n}{N}, \quad 0 \le n < N$$

and a Hanning window with 'N' points is mathematically represented by:

$$h_n(n) = 0.5(1 - \cos\frac{2\pi n}{N}), \quad 0 \le n < N$$

A window function has non-zero